B.Tech. Degree VI Semester Examination April 2013

ME 601 INSTRUMENTATION AND CONTROL SYSTEMS
(*2006 Scheme*)

Time : 3 Hours | Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

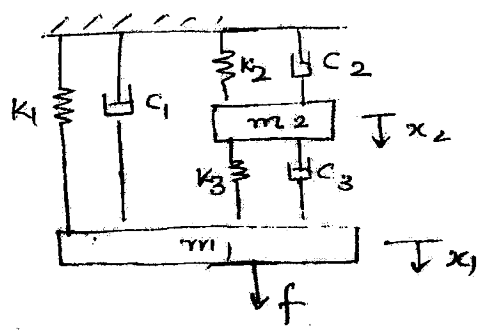
I. (a) Describe the working of liquid-in-glass thermometer as a first order instrument.

(b) Explain the terms 'static calibration' and 'sensitivity of an instrument'.

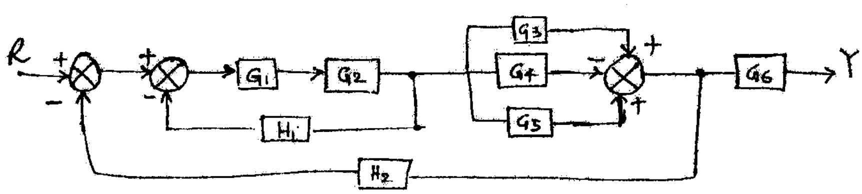
(c) Describe the working of a pyrometer.

(d) Explain the working of a Gieger Muller counter.

(e) Find the system equation of the mechanical system shown in the figure.



(f) Find the transfer function of the system shown.



(g) Explain the working of a stepper motor.

(h) Examine the stability of a system having the characteristic equation $s^3+2s^2+3s+10 = 0$.

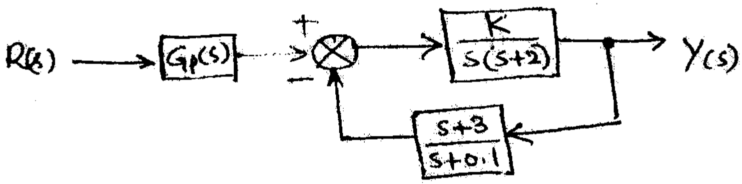
PART B

(4 × 15 = 60)

II. A system has $G(s) = \dfrac{20}{s^2+5s+5}$ and unity feed back. Determine: (i) the damping factor (ii) natural frequency (iii) damped frequency (iv) percentage overshoot (v) settling time for 2% criterion. (15)

OR

III. A feed back system is shown in the figure. (15)



(i) Determine the steady state error for a unit step when $K = 0.4$ and $G_p(s) = 1$.

(ii) Select an appropriate value for $G_p(s)$ so that the steady state error is equal to zero for the unit step input.

IV. Sketch the root locus for $G_{(s)}H_{(s)} = \dfrac{K}{s(s+3)(s+5)}$ $(k > 0)$ (15)

OR

V. A unity feed back control system has $G_{(s)} = \dfrac{40}{s(s+2)(s+5)}$. Draw the Bode plot. Also find the gain margin and phase margin. (15)

VI. (a) Explain the step response of a second order instrument. (8)

(b) Distinguish between systematic error and random error. (7)

OR

VII. What are the different methods of correcting spurious inputs to an instrument? (15)

VIII. (a) Explain the working of ORSAT apparatus. (8)

(b) Explain the working principle materials used and reference junction considerations for a thermo couple. (7)

OR

IX. (a) Describe the working of sound level meter. (8)

(b) Explain any one method of torque measurement. (7)

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